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BIOGRAPHICAL SKETCH

Dr. Liu is a Professor at the School of Control and Computer Engineering, North China Electric Power University (NCEPU). Before joining NCEPU in June 2026, he was an Associate Professor at the State Key Laboratory of Processors (SKLP), Institute of Computing Technology (ICT), Chinese Academy of Sciences (CAS). He received his Ph.D. degree from The University of Hong Kong in 2016, and his Master's and Bachelor's degrees from Harbin Institute of Technology in 2009 and 2007, respectively.

Dr. Liu's research focuses on reconfigurable and reliable architectures for AI computing systems, with particular emphasis on large-scale spatial architectures, domain-specific accelerators, and hardware–software co-design. More recently, his work has extended to LLM-driven hardware design automation, including hardware code generation, code-generation agents, and automated SoC design. His work addresses key challenges in performance scalability, energy efficiency, and system reliability, especially for data-intensive and large-model AI workloads. He has published around 100 papers in leading international conferences and journals, and has received multiple awards, including Best Paper Awards at the Great Lakes Symposium on VLSI (2021) and IEEE Transactions on Computers (2021), the CAS Outstanding Award for Technological Achievements Transformation (2019), and the Huawei Olympus Pioneering Award (2023).

Dr. Liu serves as an Associate Editor for IEEE Transactions on Emerging Topics in Computing, and as an **Area Chair for ICML 2027 and ICLR 2027**. He has served on the technical program committees of major design automation and FPGA conferences, including DAC, FPT, ATS, DFT, and DFTS, and as a reviewer for leading AI conferences including NeurIPS, ICML, ICLR, and AAAI, as well as for IEEE journals such as TCAD, TVLSI, TPDS, TC, TECS, and TRETS. He is a Senior Member of IEEE and CCF, and a member of ACM.

EDUCATION & APPOINTMENTS

Professor, North China Electric Power University	*2026-06 - now*
Associate Professor, Institute of Computing Technology, CAS	*2018-06 - 2026-05*
Research Fellow, National University of Singapore	*2016-12 - 2018-06*
Ph.D. Computer Engineering, The University of Hong Kong	*2011-09 - 2016-11*
M.E. Electronic Engineering, Harbin Institute of Technology	*2007-09 - 2009-07*
B.E. Electronic Engineering, Harbin Institute of Technology	*2003-09 - 2007-07*

ACADEMIC SERVICE

2027: Area Chair of ICML, ICLR

2024-now: Associate Editor of IEEE Transactions on Emerging Topics in Computing

2025-now: Review Board Member of IEEE Transactions on Parallel and Distributed Systems

2022-2023: TPC of DFT, FPT, CFTC, ATS, DFTS

2023: Registration Chair of ATS

2025: TPC of FPT, DFTS, DAC

Reviewer: NeurIPS, ICML, ICLR, AAAI (since 2023); TCAD, TVLSI, TC, TPDS, TECS, TRETS (since 2018)

RESEARCH GRANTS

[PI] Fault-Tolerant Chip Design under the Approximate Computing Paradigm, NSFC, 606,000¥ *2027 - 2030*

[PI] Cross-Layer Processor Optimization Infrastructures, CAS, 10,740,000¥ *2024 - 2028*

[PI] Representation and Application of Multimodal IC Data, CAS Computing Power Program, 710,000¥ *2025 - 2026*

[PI] Reliability Evaluation of Remote Sensing Systems, CASC, 500,000¥ *2024 - 2024*

[PI] Computational Storage Based Big Data Processing, MIP, 1,300,000¥ *2023 - 2024*

[PI] Resilient AI Tool Chain for COTS Devices, CASC, 300,000¥ *2023 - 2023*

[PI] Elastic Fault-Tolerant DLA Architecture, NSFC, 570,000¥ *2022 - 2025*

[PI] Energy-Efficient Graph Processing Acceleration on FPGAs, NSFC, 270,000¥ *2020 - 2022*

[PI] Automatic DLA Hardening, SKLCA, 300,000¥ *2021 - 2022*

[Co-PI] Cross-Layer Chip DSE Methods, NKRDP, 1,600,000¥ *2023 - 2025*

[Co-PI] Micro-storage Based Computational Storage, Huawei, 1,200,000¥ *2023 - 2024*

[Co-PI] Open AIoT Processor Architecture Design, CAS, 12,000,000¥ *2020 - 2021*

[Co-PI] Computation Storage Based Stream Processing, Huawei, 1,200,000¥ *2020 - 2021*

PUBLICATIONS

[1] Kaiyan Chang, Mingzhi Chen, Yunji Chen, Zhirong Chen, Dongrui Fan, Junfeng Gong, Nan Guo, Yinhe Han, Qinfen Hao, Shuo Hou, Xuan Huang, Pengwei Jin, Changxin Ke, Cangyuan Li, Guangli Li, Huawei Li, Kuan Li, Naipeng Li, Shengwen Liang, Cheng Liu, Hongwei Liu, Jiahua Liu, Junliang Lv, Jianan Mu, Jin Qin, Bin Sun, Chenxi Wang, Duo Wang, Mingjun Wang, Ying Wang, Chenggang Wu, Peiyang Wu, Teng Wu, Xiao Xiao, Mengyao Xie, Chenwei Xiong, Ruiyuan Xu, Mingyu Yan, Xiaochun Ye, Kuai Yu, Rui Zhang, Shuoming Zhang, and Jiacheng Zhao. "Large processor chip model". In: *China Science Information Science* (2026). **CCF-A**.

[2] Junfeng Gong, Long Cheng, Jiawei Nian, Cheng Liu（通信）, and Huawei Li. "MCU-MixQ: A HW/SW Co-optimized Mixed-precision Neural Network Design Framework for MCUs". In: *ACM Transactions on Embedded Computing Systems (TECS)* (2026).

[3] Junfeng Gong, Zhiyi Wei, Junying Chen, Cheng Liu (通信), and Huawei Li. "From Large to Small: Transferring CUDA Optimization Expertise via Reasoning Graph". In: *The International Conference on Learning Representations (ICLR)* (2026). **CCF-A**.

[4] Zizhen Liu, Fangzhiyi Wang, Mengdi Wang, Jing Ye, Hayden Kwok-Hay So, Cheng Liu(通信), and Huawei Li(通信). "Lifecycle Cost-Effectiveness Modeling for Redundancy-Enhanced Multi-Chiplet Architectures". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)* (2026). **CCF-A**.

[5] Xinghua Xue, Cheng Liu(通信), Feng Min, Hui Li, Kai Zhang, and Yinhe Han. "Adaptive Soft Error Protection for Neural Network Processing". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)* (2026). **CCF-A**.

[6] Jiazheng Zhang, Cheng Liu (通信), Long Cheng, Xiaowei Li, and Huawei Li. "Understanding and Mitigating Errors of LLM-Generated RTL Code". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)* (2026). **CCF-A**.

[7] Rengang Zhang, Leyan Wang, Zizhen Liu, Bin Sun, Naixing Wang, Cheng Liu, Jing Ye, and Huawei Li. "STAC: Spatial-Temporal Activation Contextualization for Resilient LLM Inference". In: *2026 ACM/IEEE Design Automation Conference (DAC)* (2026). **CCF-A**.

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[9] Mingkai Chen, Tianhua Han, Cheng Liu(通信), Shengwen Liang, Kuai Yu, Lei Dai, Ziming Yuan, Ying Wang, Lei Zhang, Huawei Li, and Xiaowei Li. "DRIM-ANN: An Approximate Nearest Neighbor Search Engine based on Commercial DRAM-PIMs". In: *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)* (2025). **CCF-A**.

[10] Guoyu Li, Chunyun Chen, Shengyu Ye, Yang Wang, Fan Yang, Ting Cao, Mohamed M. Sabry Aly, Cheng Liu, and Mao Yang. "LUT-DLA: Lookup Table as Efficient Extreme Low-Bit Deep Learning Accelerator". In: *The 31st IEEE International Symposium on High-Performance Computer Architecture(HPCA)* (2025). **CCF-A**.

[11] Ziming Yuan, Lei Dai, Wen Li, Jie Zhang, Ying Wang, Cheng Liu, Huawei Li, Xiaowei Li, Jiafeng Guo, Peng Wang, Renhai Chen, and Gong Zhang. "NeuVSA: A Unified and Efficient Accelerator for Neural Vector Search". In: *The 31st IEEE International Symposium on High-Performance Computer Architecture(HPCA)* (2025). **CCF-A**.

[12] Qing Zhang, Siting Liu, Yajuan Hui, and Cheng Liu(通信). "ApproxPilot: A GNN-based Accelerator Approximation Framework". In: *International Symposium of EDA (ISEDA)* (2025).

[13] Xinmiao Zhang, Zheng Feng, Shengwen Liang, Xinyu Chen, Lei Zhang, and Cheng Liu(通信). "Graphitron: A Domain Specific Language for FPGA-based Graph Processing Accelerator Generation". In: *The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES)* (2025).

[14] Xinmiao Zhang, Cheng Liu(通信), Shengwen Liang, Hayden Kwok-hay So, Ying Wang, Lei Zhang, Huawei Li, and Xiaowei Li. "TaijiGraph: An Out-of-core Graph Processing System Enhanced with Computational Storage". In: *The 39th IEEE International Parallel & Distributed Processing Symposium (IPDPS)* (2025).

[15] Xinmiao Zhang, Cheng Liu(通信), Shengwen Liang, Chengwei Xiong, Yu Zhang, Lei Zhang, Huawei Li, and Xiaowei Li. "Frontier-guided Graph Reordering". In: *The 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP)* (2025). (Poster).

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[17] Xinmiao Zhang, Xiaorong Qin, Lei Zhang, and Cheng Liu(通信). "HisOrder: A Historical Frontier-Aware Graph Reordering for Efficient BFS". In: *The 1st Workshop on FastCode Programming Challenge (FCPC) in PPoPP* (2025).

[18] Mingkai Chen, Cheng Liu(通信), Shengwen Liang, Lei He, Ying Wang, Lei Zhang, Huawei Li, and Xiaowei Li. "An Energy-Efficient In-Memory Accelerator for Graph Construction and Updating". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (2024). **CCF-A**.

[19] Long Cheng, Yan Gu, Qingzhi Liu, Lei Yang, Cheng Liu, and Ying Wang. "Advancements in Accelerating Deep Neural Network Inference on AIoT Devices: A Survey". In: *IEEE Transactions on Sustainable Computing* (2024).

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[28] Pre-order Softcover Book. "Built-in Fault-Tolerant Computing Paradigm for Resilient Large-Scale Chip Design A Self-Test, Self-Diagnosis, and Self-Repair-Based Approach". In: (2023).

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[31] Chengsi Gao, Ying Wang, Cheng Liu, Mengdi Wang, Weiwei Chen, Yinhe Han, and Lei Zhang. "LayerPuzzle: Allocating and Scheduling Multi-task on Multi-core NPUs by Using Layer Heterogeneity". In: *2023 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE. 2023.

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[37] Neo Wei Ming, Zhehui Wang, Cheng Liu, Rick Siow Mong Goh, and Tao Luo. "MA-BERT: Towards Matrix Arithmetic-only BERT Inference by Eliminating Complex Non-linear Functions". In: *The 11th International Conference on Learning Representations (ICLR)*. 2023.

[38] Xinghua Xue, Cheng Liu(通信), Haitong Huang, Bo Liu, Ying Wang, Bing Yang, Tao Luo, Lei Zhang, Huawei Li, and Xiaowei Li. "ApproxABFT: Approximate algorithm-based fault tolerance for vision transformers". In: *arXiv preprint arXiv:2302.10469* (2023).

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[52] Xinghua Xue, Haitong Huang, Cheng Liu(通信), Tao Luo, Lei Zhang, and Ying Wang. “Winograd convolution: A perspective from fault tolerance”. In: *Proceedings of the 59th ACM/IEEE Design Automation Conference* (2022). **CCF-A**.

[53] Xiandong Zhao, Ying Wang, Cheng Liu, Cong Shi, Kaijie Tu, and Lei Zhang. “Network Pruning for Bit-Serial Accelerators”. In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (2022). **CCF-A**.

[54] Weiwei Chen, Ying Wang, Gangliang Lin, Chengsi Gao, Cheng Liu, and Lei Zhang. “CHaNAS: coordinated search for network architecture and scheduling policy”. In: *Proceedings of the 22nd ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems*. 2021.

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